

PROJECT DOCUMENTATION

IoT Weed Detection & Removal Robot

Backend API Documentation

Prepared for: Member 1 (Backend Developer)
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1. Overview

This document defines the REST API contract between the ESP32-based sensor / control firmware and the backend server for the IoT Weed Detection & Removal Robot. It lists every available endpoint, the expected request payload, and the purpose of each call so that frontend, firmware, and backend teams can integrate against a single shared reference.

- ▶ 14 endpoints covering status, control, movement, and cutter operations
- ▶ Two HTTP methods in use: GET for read-only state, POST for commands and updates
- ▶ All request and response bodies are JSON

2. Endpoint Reference

The table below lists every route exposed by the backend, grouped in the order a client typically calls them.

Method	Endpoint	Request Body	Purpose
GET	/api/status	None	Current robot / sensor status
GET	/api/robot/status	None	Complete robot state
GET	/api/logs	None	Robot activity logs
GET	/api/sessions	None	Last five sessions
POST	/api/update	{"weed":true,"moisture":60}	ESP32 sensor update
POST	/api/control	{"autoMode":true,"removal":false}	Update control settings
POST	/api/movement/forward	None	Move forward
POST	/api/movement/backward	None	Move backward

Method	Endpoint	Request Body	Purpose
POST	/api/movement/left	None	Move left
POST	/api/movement/right	None	Move right
POST	/api/movement/stop	None	Stop robot
POST	/api/robot/mode	{"mode": "AUTO"} / {"mode": "MANUAL"}	Change robot mode
POST	/api/cutter/start	None	Start cutter
POST	/api/cutter/stop	None	Stop cutter

Legend

- GET — Read-only request, no side effects, no request body
- POST — Sends or changes state on the robot / server

3. Example Response

GET /api/robot/status returns the complete current state of the robot in the following shape:

```
{
  "movement": "BACKWARD",
  "mode": "MANUAL",
  "cutter": true,
  "weedStatus": "Weed detected",
  "moisture": 60,
  "scansToday": 1,
  "weedsDetected": 1,
  "weedsRemoved": 0,
  "battery": 80,
  "esp32Connected": false
}
```

Note: *esp32Connected* reflects live link health with the sensor unit; treat *weedStatus* and *moisture* as the most recent values reported by the last */api/update* call.